

StateM: Reaching 95.3% Raw Accuracy, or a \$15 Frontier Run, on Terminal-Bench 2.1 via Harness Scaling

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Somewhere on the Earth ¹

Abstract

Long-horizon agents can fail even when their underlying models can solve the constituent steps. They may lose track of mutable state, fail to reactivate lessons from earlier executions, skip known procedures, or stop prematurely. We bet on *harness scaling* to improve the execution system around an agent without changing its model weights. We introduce STATEM, an agent-native runtime that organizes execution around durable states, phase-local context, checked transitions, recoverable runbooks, and versioned procedural practices that agents and users can inspect together.

On Terminal-Bench 2.1, GPT-5.5 xhigh with STATEM reaches 92.1% accuracy, compared with an 83.1% reference and surpassing GPT-5.6 Sol Ultra (91.9%). The same runbook transfers to GPT-5.6 without modification. With GPT-5.6 Sol xhigh, the system records **95.3% raw accuracy** across 445 public-submission trials and succeeds at least once on each of the 89 tasks. The frozen profile also raises GPT-5.6 Luna from 76.7% to 85.4%, numerically above the 84.9% Sol xhigh reference.

Next, starting from the same runtime, runbook structure, and golden rules, it takes less than \$38 of adaptation cost to raise DeepSeek-V4 Flash from 82.7% to 88.1% on the full benchmark under standard timeouts. It reaches 89.1% on a disclosed 88-task common core. Extending the timeout only for the remaining latency-sensitive task yields a descriptive full-suite aggregate matching the reported 88.8% GPT-5.6 Sol max result. The complete DeepSeek final-score evidence costs approximately **\$15 in API usage**, compared with the \$574.68 reported for the GPT reference. All recorded DeepSeek adaptation and evaluation API expenditure totals \$52.22.

On BusinessBench, we develop family-specific runbooks on separate development sets using the same harness-development principles. The first frozen held-out evaluation yields smaller aggregate gains of 0.55 macro and 1.34 micro percentage points, while two mechanism-matched task families improve by 10.04 points. These results suggest that concrete runbook rules generalize when tasks share the relevant execution structure, while the methodology for identifying and enforcing such controls remains applicable across more heterogeneous workflows. STATEM converts selected generalizable postmortem findings into persistent, executable preconditions and practices. It thereby makes explicit and enforceable what the agent should know, what it should do, and which learned practices should be reactivated in future runs — all through stateful controls. Core code and several runtime cases is open sourced at [github](#).

1 Introduction

Long-horizon agents often fail in a revealing way: the underlying model appears capable of solving each local step, yet the complete run still fails. The agent deviates from its plan, loses track of mutable task state, skips a necessary check, repeats an unproductive action, or stops before the requested deliverable is verifiably

¹This work was conducted in the authors’ *personal time* and does NOT reflect the views of any affiliated organization. Ziheng and Yaxin are core contributors (*i.e.*, equally-first authors).

complete (Yao et al., 2024; Barres et al., 2025; Deng et al., 2025; Zhu et al., 2026; Liu et al., 2026). These failures matter because real workflows and agent benchmarks evaluate the completed job, not whether the model knew how to perform most of its constituent steps.

The dominant response to such failures is to improve the model: scale pretraining, add post-training data, increase test-time reasoning, or introduce additional agents. We study an orthogonal question:

How much apparent model failure is actually failure of the harness that maintains state, constrains execution, verifies progress, and recovers from errors?

We call the systematic improvement of this surrounding control layer *harness scaling*. Harness scaling does not replace model scaling. It asks whether more of a model’s existing capability can be converted into completed, reliable work by improving the runtime around it. This leads to three progressively stronger empirical tests: *first*, can a better harness improve a fixed model without changing its weights? *Second*, can a harness developed with one model transfer to a newer model without retuning? *Third*, can the resulting control principles improve performance beyond the benchmark on which they were developed?

Why external control?. We motivate harness scaling through two design hypotheses. First, *control-signal dilution* can arise when a compact plan and its completion criteria are surrounded by an increasingly long trace of commands, observations, and repairs. Second, *mutable-state ambiguity* can arise when completed goals, pending dependencies, failed attempts, and valid next actions must be reconstructed from an append-only history rather than read from an authoritative current state. We treat these as operational hypotheses, not as fundamental claims about transformer attention. Both suggest externalizing procedural state, refreshing the control information relevant to the current phase, and checking evidence before important transitions.

A control-layer design space. Existing systems expose a tradeoff between runtime enforceability and agent autonomy. State-machine and graph-based runtimes provide explicit workflow state, persistence, conditional routing, and recovery (Wu et al., 2024; Wang and Duan, 2024), but typically organize execution around a developer-authored controller. General-purpose CLI agents preserve a broader reasoning and action space, but their plans, instruction files, memories, and hooks do not by themselves form a unified, transition-aware control surface. Figure 1 summarizes these typical operating points. The distinction is not whether a system has state, but who can inspect, modify, and operate the stateful control layer during execution.

A complementary line of work studies reliability over even longer timescales. AgingBench shows that an agent with frozen weights remains a changing system as its memory is compressed, retrieved, revised, and maintained (Zhu et al., 2026). Its typed-state and runtime-control interventions further indicate that some longitudinal failures require explicit state rather than additional text context alone. Related work and the precise boundaries of our contribution are discussed in Section 2.

StateM: an agent-native control layer. We introduce STATEM, a lightweight runtime for long-running CLI agents. Its control layer takes the form of a human-readable YAML runbook containing states, valid transitions, state-local instructions, hooks, checks, and recovery rules. Through ordinary command-line operations, the agent can inspect its current state, request a transition, examine failed conditions, review execution history, recover after an interruption, and update permitted runbook artifacts. A human supervisor can inspect, edit, version, and audit the same runbook.

STATEM follows two principles. First, each state is a *context-and-contract boundary*. Entering a state refreshes the active instructions and durable task information for that phase. Leaving the state requires the agent to satisfy explicit exit conditions. The host runtime can evaluate executable conditions directly. Conditions that require semantic judgment are recorded as auditable attestations and may trigger further review or repair. This distinction matters because an agent’s declaration of completion does not constitute independent verification. Second, the agent and user *share* the control layer. The agent remains the primary executor and retains a unified reasoning loop, but its progress is exposed through a control surface that the host runtime can enforce and the user can revise. STATEM is agent-native in a specific sense: the agent operates the control layer through the same CLI action space it uses to perform the task. The runtime requires no model modification, fine-tuning, or access to model internals.

STATEM combines three properties that are not usually available in one system:

Design space of control layers for long-horizon agents

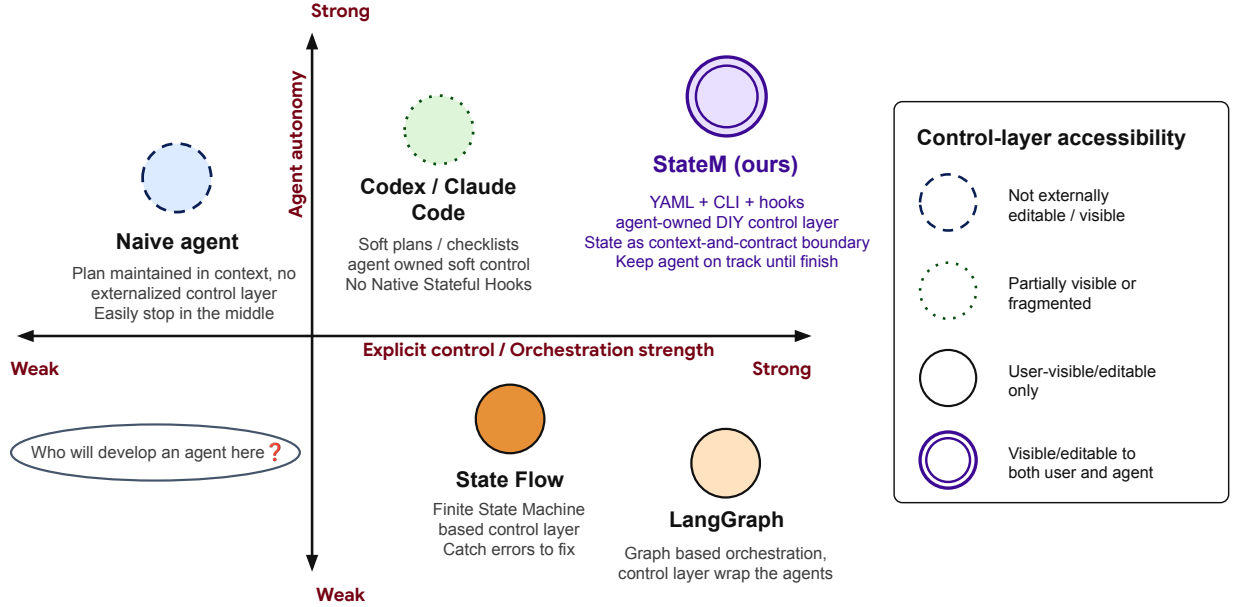


Figure 1 Conceptual design space for agent control layers. The vertical axis denotes *runtime enforceability*: the extent to which the runtime maintains authoritative execution state, constrains valid transitions, and can block incomplete handoffs. The horizontal axis denotes *agent autonomy*: the latitude retained by the primary agent to choose tools, revise procedures, and operate over a broad action space. Marker style denotes whether the control artifact is primarily developer-owned, agent-editable, or jointly editable by the agent and user. Positions represent typical operating abstractions rather than architectural limits or empirical performance rankings. StateM targets the upper-right region through a shared runbook that combines broad agent autonomy with explicit, enforceable state transitions. Evaluation and harness-optimization methods, including AgingBench and automated harness adaptation, are discussed in the text but are not plotted because they operate on different dimensions.

1. state boundaries provide both refreshed control contexts and explicit transition contracts;
2. a general-purpose agent remains the primary executive instead of being decomposed into a fixed collection of node-local model calls; and
3. the runbook is a shared, machine-operable artifact that agents and users can inspect, audit, and revise through the same execution environment.

This representation allows the harness itself to be optimized. After a run, failures can be classified as missing context, invalid transitions, weak checks, premature handoffs, ineffective recovery, or other control-layer defects. The working agent or a separate hyper-agent can then propose changes to state boundaries, prompts, hooks, checks, recovery rules, and practice activation conditions. Candidate changes are reviewed, tested for regressions, and incorporated into a new runbook version. The model weights remain unchanged, while selected procedural knowledge accumulates in the external control layer.

This process addresses a failure mode that arises across executions rather than within a single run. An agent may correctly diagnose a failure but neither retain nor invoke the resulting lesson when the same risk recurs. We call this the *procedural-memory gap*. It complements an *epistemic gap*, in which relevant knowledge or an appropriate method is unavailable at the decision point, and a *procedural-compliance gap*, in which the correct procedure is active but remains incomplete or not completely followed by the agent. State-local context, versioned practices, and checked transitions target these three gaps, respectively. A state records the position of the current run. Procedural memory records what earlier runs have taught the system to do at that point.

We call the resulting development process *failure-driven harness optimization*. It is related to recent work on improving smaller models through stronger or automatically adapted harnesses (Yang et al., 2026). STATEM

focuses on a machine-operable, agent-native runbook representation. We use this representation to study which layers of procedural control transfer across model versions, providers, and task families.

Results and scope. We organize the experiments around two complementary frontiers, summarized in Fig. 5. At the quality frontier, GPT-5.5 xhigh with STATEM reaches 92.1% accuracy on Terminal-Bench 2.1, compared with the 83.1% GPT-5.5 reference. Across five trials, the system solves 88 of the 89 tasks at least once. Its 92.1% mean is also numerically close to the 91.9% next-generation reference. This comparison illustrates that the observed system-level harness gain can resemble a model-generation upgrade. We then freeze the runbook developed with GPT-5.5 and apply it to GPT-5.6 without modifying its control logic. With GPT-5.6 Sol xhigh, the combined system records 95.28% raw accuracy in a public Terminal-Bench 2.1 submission. This result corresponds to 424 successful trials out of 445, with at least one successful trial on every task.¹ The frozen runbook also raises GPT-5.6 Luna from 76.7% to 85.4%, a gain of 8.7%. This result is numerically above the 84.9% Sol xhigh reference.

The cost frontier has a transfer boundary. Starting from the same runtime, high-level runbook structure, and golden rules, less than \$38 of DeepSeek API expenditure is sufficient to adapt the concrete practices. The adapted system reaches $392/445 = 88.09\%$ on the full 89-task benchmark under standard timeouts. On the disclosed 88-task common core, which excludes the latency-sensitive `gpt2-codegolf` task, it reaches $392/440 = 89.09\%$. Extending the timeout only for that task yields a descriptive full-suite aggregate above the reported 88.8% GPT-5.6 Sol max result. The complete DeepSeek final-score evidence costs approximately \$15 in realized API charges, compared with \$574.68 reported for the GPT-5.6 Sol max reference. All recorded DeepSeek API expenditure across profile adaptation and final evaluation is \$52.22. These amounts are approximately $38.9\times$ and $11.0\times$ lower, respectively.

We also evaluate task-side generalization on BusinessBench (Yang et al., 2026) using Codex with GPT-5.6 Luna. The dataset contains 477 eligible instances across seven task families. Of these, 405 instances across six families receive a STATEM intervention. In the first frozen one-shot evaluation, held-out performance changes from 84.67% to 85.22% under an equal-family macro average, and from 84.44% to 85.78% under an instance-weighted micro average. Development performance improves from 86.07% to 91.71%, while performance across all treated instances improves from 84.76% to 88.72%. An exploratory two-family slice containing Budget Approval and Machine Operating improves from 71.91% to 81.94%. This pattern suggests that the largest gains occur when the control profile matches a specific execution boundary.

Contributions. This work makes five contributions:

- We formulate *harness scaling* as a capability axis complementary to model scaling. We distinguish within-model improvement, frozen cross-generation transfer, adapted cross-provider transfer, and held-out task generalization as separate empirical tests.
- We introduce STATEM, an agent-native runtime in which durable states provide refreshed control contexts and explicit transition contracts while preserving the unified reasoning loop of a general-purpose CLI agent.
- We operationalize three sources of execution failure: epistemic gaps, procedural-memory gaps across runs, and procedural-compliance gaps within a run. State-local context, versioned practices, and checked transitions provide corresponding control points.
- We develop a failure-driven optimization loop that converts selected execution failures into versioned changes to prompts, hooks, checks, recovery rules, state boundaries, and practice activation conditions. This process allows procedural knowledge to accumulate outside the model weights.
- We evaluate the system across GPT-5.5, two GPT-5.6 variants, DeepSeek-V4 Flash, Terminal-Bench 2.1 and BusinessBench. The evaluation covers raw-score adjudication sensitivity, frozen and adapted transfer, task-level regressions, realized API expenditure, and explicit disclosure of benchmark adaptation.

¹**Evaluation status as of August 11, 2026.** The 95.28% figure is the raw, pre-adjudication accuracy computed by the Terminal-Bench 2.1 submission pipeline in PR #142: <https://github.com/harbor-framework/terminal-bench-2-1/pull/142>. The submission passed ten automated checks covering task digests, standard runtime settings, trial coverage, and trajectory availability. The PR remains open and has not been merged into the leaderboard. We agree that four rewarded trajectories identified during review should not count. Scoring those trials as zero gives $420/445 = 94.38\%$. Scoring all nine trajectories currently flagged for possible reward hacking as zero gives $415/445 = 93.26\%$. We therefore report 95.28% only as a raw public submission score. The submission pipeline reports 1.178 billion tokens and \$1,062.95 in model cost. Five-trial task coverage does not measure single-run reliability.

The results do not suggest that model scaling has ceased to matter or that a static runbook can dominate every model, task, and benchmark. We instead advocate that runtime semantics and principles for harness development can transfer broadly, while concrete practices transfer locally or require inexpensive adaptation. Model capability and execution reliability make separate contributions to system performance. In the settings studied here, **the model appears not to be the (main) bottleneck**.

2 Related Work

StateM intersects four lines of work: long-horizon planning, stateful agent orchestration, long-lived agent memory, and automated harness adaptation. Its novelty does not lie in any one component in isolation. Instead, it lies in their combination through an agent-native, jointly editable execution representation.

Planning and long-horizon agent execution. Planning is a standard response to long-horizon complexity: an agent first decomposes a task and then executes the resulting procedure. Nevertheless, current agents frequently depart from their plans, omit necessary steps, or terminate without satisfying the original completion criteria (Liu et al., 2026). Plans represented only in natural language remain advisory unless the runtime maintains progress and checks whether required conditions have been met.

At the same time, stronger orchestration is not universally beneficial. Decomposing a procedure into many separately prompted nodes can fragment reasoning and introduce routing errors. Recent evidence finds settings in which a strong model given the full procedure in context outperforms a more heavily orchestrated version of the same procedure (Dennis et al., 2026). StateM is motivated by this tension. It seeks stronger execution control without decomposing the primary agent into a sequence of narrowly scoped model calls.

Stateful and graph-based orchestration. Finite-state workflows and graph-based agent runtimes already provide important control mechanisms. StateFlow represents problem solving through explicit states and transitions, demonstrating that state-driven execution can improve task grounding and failure recovery (Wu et al., 2024). LangGraph and related durable workflow systems provide persistent state, conditional edges, checkpointing, interruption, and human-in-the-loop recovery (Wang and Duan, 2024). StateM does not claim to introduce these capabilities.

The difference is the default execution abstraction. In a conventional controller-led graph, a developer specifies the workflow and invokes models or agents within its nodes. The graph may be highly flexible, but the primary control artifact remains external to the executing agent. StateM instead keeps a general-purpose CLI agent as the primary executive and exposes the control layer within its ordinary tool environment. The agent can inspect its current state, request transitions, examine failed conditions, and propose permitted runbook changes without leaving its normal action space.

This distinction should not be read as a strict separation between architectures. A graph runtime can host a highly autonomous agent, and a StateM runbook could be implemented on top of a durable workflow engine. Figure 1 describes typical design choices rather than hard capability boundaries. The specific contribution of StateM is a lightweight agent-facing control representation in which state boundaries are both refreshed control contexts and auditable transition contracts.

CLI agents, soft plans, and user control. General-purpose CLI agents such as Codex and Claude Code preserve broad autonomy over tool selection, file manipulation, code execution, and iterative problem solving. They also expose useful control fragments, including planning prompts, TODO lists, rule files, memory files, and host-side hooks. These mechanisms make the agents highly adaptable, but they do not necessarily establish one authoritative execution state. Plans and checklists are usually soft natural-language artifacts, while hooks are commonly attached to tool or lifecycle events rather than to semantically meaningful transitions.

StateM organizes these fragments into a shared runbook containing current state, valid transitions, state-local instructions, checks, hooks, recovery rules, and execution history. The user can therefore intervene by editing the same control artifact that the agent and runtime operate, rather than only by adding another natural-language correction to the interaction history. This shared ownership is the main distinction represented by the third visual encoding in Figure 1.

Long-horizon agents and agent aging. Long-horizon execution within one task is related to, but distinct

from, reliability across many sessions. AgingBench frames deployed agents as time-evolving systems whose effective behavior can change even when model weights remain frozen (Zhu et al., 2026). It identifies four longitudinal failure mechanisms: compression aging, interference aging, revision aging, and maintenance aging. It further localizes failures to writing, retrieval, utilization, and lifecycle handling through temporal dependency graphs and counterfactual diagnostic probes.

AgingBench is particularly relevant to StateM in two ways. First, its revision aging results show that mutable or derived state can fail even when additional text memory is available, supporting the need for explicit state representations. Second, AgingBench studies a typed-state overlay and a threshold-triggered runtime controller as targeted interventions. These results connect diagnosis to runtime control, but at a different level from StateM. The AgingBench interventions maintain selected memory variables or activate memory-policy changes across sessions. StateM provides a general procedural runtime for representing phases, transitions, evidence, checks, and recovery within open-ended agent work.

The two approaches are therefore complementary. AgingBench asks how reliability changes over an operational lifespan, which mechanism is degrading, and where a repair should target. StateM asks how an agent should represent and enforce its current procedural state while completing a long-running task. Combining lifespan diagnostics with StateM runbook adaptation is a natural direction for future work, but StateM does not by itself solve compression, retrieval, or maintenance aging.

Harness adaptation and self-improvement. A growing body of work shows that agent performance depends strongly on the runtime harness and that the harness can itself be optimized. Agentic Harness Engineering uses execution observability to evolve coding-agent tools, middleware, prompts, and memory from trajectory feedback (Lin et al., 2026; Gu, 2026). Life-Harness converts recurring failures into reusable interventions over environment contracts, procedural skills, action realization, and trajectory regulation, then freezes the resulting harness for held-out evaluation (Xu et al., 2026). Self-Harness lets an agent mine its weaknesses, propose harness modifications, and promote them through regression-aware validation (Zhang et al., 2026). Better Harnesses, Smaller Models studies automated harness adaptation as a way for lower-cost models to recover much of the performance of larger models (Yang et al., 2026).

These works make two points clear. First, model identity alone is not a sufficient description of an agent system. Second, neither harness adaptation nor cross-model harness transfer is, by itself, a novel claim. StateM differs primarily in the artifact being optimized and operated. Its harness is an explicit state-machine runbook whose states, transition conditions, hooks, repair paths, and history are directly available to the executing agent and human supervisor. The same representation serves three roles: runtime control, an audit surface, and the search space for failure-driven improvement.

StateM is thus closest to prior harness-evolution methods in its hyper-agent loop, but its central contribution is the execution substrate that this loop modifies. Unlike prompt-only optimization, StateM can alter transition-time checks, recovery behavior, and state-local context. Unlike a fully external workflow graph, the resulting runbook remains inside the primary agent’s normal tool environment. Unlike longitudinal diagnostic benchmarks, it acts directly on within-run procedural execution.

Positioning StateM. The quadrant in Figure 1 summarizes the resulting design claim. Soft plans and conventional CLI control fragments preserve autonomy but offer limited transition-level enforcement. Developer-authored workflow graphs offer strong control but typically place the agent within an externally specified execution structure. StateM targets the less explored combination of broad agent autonomy, enforceable state transitions, and a control artifact jointly visible and editable to the agent and user. This positioning does not imply that one point in the quadrant is universally superior. Stable, repetitive workflows may be best served by a fixed graph, while short or exploratory tasks may need little control beyond an instruction file. StateM is intended for the intermediate regime: tasks that are open-ended enough to require a general-purpose agent, but long and consequential enough that soft plans and self-declared completion are insufficient.

3 StateM: Agent-Native Control for Long-Horizon Execution

Section 1 identified two operational pressures in long-horizon execution: the control signal carried by a plan can weaken as the trace grows, and the current task state can become ambiguous when it must be

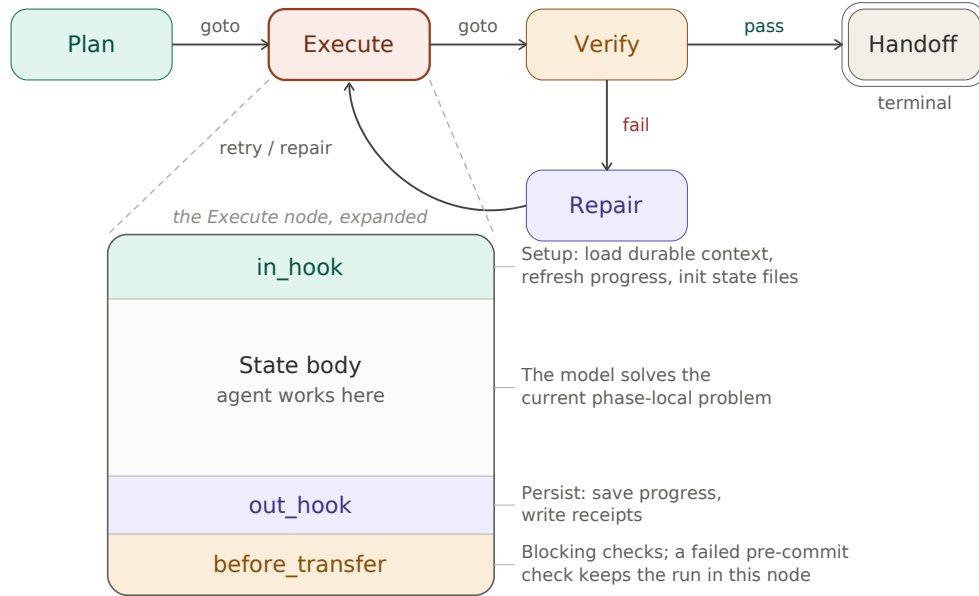


Figure 2 The StateM control surface. A runbook is a directed state machine over coarse phases (top). Each `goto` follows a checked and logged transition protocol; verification failures can route execution through a configured repair state, and successful completion is recorded only at a terminal state. Expanding a node (bottom) reveals its phase-local structure: an `in_hook` for setup and context loading, a state body in which the model performs the open-ended work, an `out_hook` for persistence, and a `before_transfer` block containing checks that must pass before the transition can commit.

reconstructed from an append-only history. Section 2 further identified a design tension between runtime enforceability and agent autonomy. StateM addresses this tension by preserving broad agent discretion *within* phases while making progress *between* phases explicit, persistent, and checkable.

The design has four requirements. First, the control layer should represent coarse phases of work without decomposing the agent into a sequence of micro-actions. Second, it should refresh phase-relevant instructions and durable progress when a phase begins. Third, it should expose explicit control points where incomplete work, failed verification, or external blockers can prevent a handoff. Finally, it should be directly operable by the agent and auditable by the user through their shared workspace.

StateM realizes these requirements as a YAML-configured state-machine runtime with a command-line interface. A versioned *runbook* defines states, transitions, prompts, hooks, conditions, and checks. The agent operates this runbook through ordinary CLI commands, while the runtime maintains execution state outside the model context. The control layer is therefore externalized from the interaction history without being hidden from either the agent or the user.

3.1 Runtime and Control Profile

StateM separates two layers that should not be conflated. The *runtime* provides the generic mechanisms for state persistence, transition validation, hook execution, history, and recovery. The *control profile*, represented by a runbook, specifies the phases, instructions, checks, and repair policies used for a particular class of work.

This distinction is important for interpreting the experiments. The StateM runtime is designed to be reusable across agents and workflows, but the content of a runbook may encode workflow-specific or experience-derived procedural knowledge. In particular, the Terminal-Bench results in Section 4 evaluate the combined runtime and an evolved, benchmark-adapted control profile. They should not be interpreted as isolating the effect of the state-machine abstraction alone.

The base runbook is a static, shareable, and versionable artifact. Multiple agents and executions can use the

same runbook, while each execution maintains separate mutable runtime state. Runbooks may be updated between versions, and permitted run-local additions can be recorded during execution, but such changes remain distinct from the underlying runtime.

3.2 States as Context-and-Contract Boundaries

The central StateM abstraction is a *phase-level state*. A state represents a meaningful stage of work rather than a single model call or tool action. A coding runbook, for example, may include states for planning, implementation, task-contract checking, self-review, repair, and handoff. The agent remains free to reason, invoke tools, edit files, and iterate within each state.

A state serves first as a *context boundary*. When the agent enters a state, StateM exposes the current phase, valid outgoing transitions, state-local instructions, and relevant durable progress. The `in_hook` performs the configured entry procedure. It may inject a prompt, run setup code, initialize state-local files, or load a compact progress record. This creates a fresh control anchor without requiring the agent to infer its current phase and outstanding obligations entirely from the preceding terminal trace.

This mechanism does not erase the model’s prior context or guarantee lossless compaction. Rather, it makes the authoritative phase and its local obligations explicit and recent. It can repair a missing-context problem when the necessary information is available to the runbook, but it cannot supply task knowledge that neither the model nor the accessible environment possesses.

A state also serves as a *contract boundary*. The state prompt describes the work expected within the phase, while exit hooks and checks specify the conditions for leaving it. The `out_hook` can persist progress, update receipts, or prepare artifacts for the next phase. The `before_transfer` block evaluates the configured exit conditions before StateM commits a transition.

StateM supports several types of checks with different evidentiary strength:

- `command` and `predicate` checks are evaluated by the host and can provide independently reproducible evidence, subject to the correctness of the command or predicate;
- `manual` checks require an explicit user or operator decision;
- `checklist` and `message` checks require the agent to acknowledge specified obligations, but remain forms of structured self-attestation; and
- `llm_review` checks add a separate semantic judgment, but do not constitute deterministic verification.

This distinction prevents a receipt or model declaration from being treated as proof merely because it appears in a structured field. StateM makes such claims visible and auditable; their reliability still depends on the mechanism that produced and verifies them.

Together, state entry and state exit address different failure modes. Entry hooks refresh the phase-local control signal and make durable progress available. Exit checks create intervention points for procedural requirements, such as running a consumer command, executing tests, recording evidence, obtaining approval, or updating a receipt. Failed checks remain visible rather than silently propagating into later phases.

3.3 Runbooks, Edges, and Transition Verification

Formally, a runbook defines

$$\mathcal{B} = (\mathcal{S}, s_0, \mathcal{S}_T, \mathcal{E}, \Phi),$$

where $\mathcal{S}$ is the set of phase-level states, s_0 is the initial state, $\mathcal{S}_T \subseteq \mathcal{S}$ is the set of terminal states, and $\mathcal{E} \subseteq \mathcal{S} \times \mathcal{S}$ is the set of permitted transitions. The state specification $\Phi(s)$ contains the phase-local prompt, entry and exit hooks, transfer checks, and references to state-local artifacts. A complete example appears in Appendix A.

An edge may additionally contain a guard or transfer hook. These edge-level conditions select among valid next states using information such as check results, receipts, previous failures, external readiness, or user-blocking status. After verification, for example, a runbook may route to `repair` when tests fail, to `handoff` when

sufficient evidence has been recorded, or to `wait` when an external service or human approval is required. Repair routing is explicit in the graph rather than an implicit instruction that the agent must remember.

All phase changes use the core operation `goto`. When the agent requests `goto TARGET`, StateM executes an ordered transition protocol:

1. verify that the requested edge from the current state to `TARGET` exists;
2. evaluate the current state's `before_transfer` checks;
3. run the current state's configured persistence or `out_hook` operations;
4. evaluate edge guards and edge-specific transfer hooks;
5. commit the target state and append the transition event to history only if all required pre-commit steps succeed; and
6. create the target-state entry and execute its `in_hook`.

If a required pre-commit check or hook fails, the run remains in the source state and records the failure, allowing the agent to inspect the unmet condition, repair the underlying problem, and retry. If the transition succeeds, StateM creates a new state-entry record and exposes the target state's instructions and obligations.

We describe this protocol as *checked, logged, and recoverable*, rather than fully transactional. StateM delays the runtime-state commit until required pre-commit operations succeed, but it cannot generally roll back arbitrary external side effects produced by hooks.² Recovery refers to the StateM execution record and configured repair procedure, not ACID rollback of the external world.

3.4 Per-Run State and Recovery

A runbook describes reusable control logic, while each execution has its own runtime record. For each run, StateM stores a run identifier, current state, current state-entry identifier, transition history, hook and check outcomes, timestamps, and references to state-local evidence files. Separating this mutable record from the runbook allows one control profile to support multiple agents or concurrent runs without conflating their progress.

This persistent record also provides a recovery anchor. After a process restart, context refresh, or model-side compaction, the agent can query StateM for the current state, prior transitions, unresolved checks, and recorded evidence. It need not reconstruct the workflow solely from a long terminal transcript. A restarted agent can therefore resume from an explicit phase and a durable set of obligations.

Recovery has a precise scope. StateM can restore its recorded control state and re-execute configured recovery steps. It cannot reconstruct work that was never persisted, restore hidden model context, or automatically reverse arbitrary external actions. Reliable recovery consequently depends on placing persistence hooks at appropriate phase boundaries and designing repeated hooks to tolerate retry.

A run may be in one of three operational situations. It may be active in a non-terminal state, paused because an external condition is unresolved, or complete in a terminal state. Reaching a non-terminal pause is not treated as successful completion. This distinction allows StateM to separate genuine handoff from temporary inability to proceed.

3.5 Shared Control, Runtime Checks, and Stop Hooks

StateM is agent-native because the executing agent operates the control layer through the same CLI environment in which it performs the task. The runbook is a normal workspace artifact rather than hidden controller code. The agent can inspect its current state, follow configured transitions, examine failures, and propose runbook changes. The user can read, edit, review, and version the same artifact. This shared surface preserves broad within-state autonomy while making cross-state obligations explicit.

²Hooks execute with the permissions of the surrounding host environment. Commands that modify external services or non-versioned artifacts should therefore be idempotent where possible and should record sufficient information for explicit compensation or retry. StateM does not make untrusted hook code safe; sandboxing and permission control remain host responsibilities.

Shared ownership does not mean unrestricted self-modification. StateM distinguishes among the versioned base runbook, run-local additions, and host-owned constraints. During execution, an agent may register additional dynamic checks when the runbook permits it. Such checks are useful when the agent discovers a risk or requirement that was not known when the runbook was written. The addition is recorded in the run history so that the user can inspect it and decide whether it should be promoted into a later runbook version.

Adding a stricter run-local check is different from weakening or deleting an existing requirement. Changes to user-owned invariants, permissions, or blocking checks should require a privileged policy decision. YAML editability alone does not provide this security boundary; deployments must define which parts of a runbook the agent may modify. In the reported cross-model transfer experiment, the evaluated base runbook is frozen so that runbook evolution is not conflated with evaluation-time adaptation.

StateM also supports host-level stop-hook integration for Codex and Claude Code. When the host receives a request to stop, the hook can inspect the current StateM status. If the run is in a terminal state or explicitly blocked on an external dependency, the stop can proceed. Otherwise, the hook returns the current phase and unmet obligations to the agent and requests continued work. This mechanism reduces premature termination and supports longer unattended runs.

A stop hook does not guarantee eventual success or termination. The agent may remain unable to satisfy a check, exhaust its model or environment budget, or repeat an ineffective repair. Stop-hook behavior should therefore remain subject to retry, time, and resource limits. Its benefit is greater execution persistence and a more explicit stopping condition, not greater reasoning capability by itself.

3.6 Scope of the Control Guarantee

StateM provides control points, not a correctness oracle. Its guarantees are strongest when a transition condition can be checked independently by a command, predicate, or human decision. Semantic self-review and agent-authored receipts remain fallible. Similarly, StateM can prevent a configured requirement from being silently skipped, but it cannot detect a missing requirement that neither the runbook nor the agent introduces.

The framework should therefore be understood as an execution substrate with three roles. It is a runtime that maintains procedural state and controls transitions; an audit surface that exposes progress, evidence, and failure; and an optimization target through which experience-derived procedural knowledge can be preserved outside model weights. The empirical gains in the following section arise from the combined runtime and control profile. The next section examines how that profile was developed, frozen, transferred across model versions, and evaluated.

4 Harness Scaling in Practice: Two Frontiers and a Hierarchy of Transfer

Harness scaling treats agent capability as a property of both model weights and the execution system that preserves state, reactivates experience, checks progress, and supports repair. Unless otherwise stated, a STATEM system comprises the base agent, the generic STATEM runtime, and an identified control profile. The results measure this combined system, including the profile’s evolved prompts, routing, checks, and practices; they do not isolate the state-machine runtime from the procedural content encoded in the runbook.

Across three axes, we study four empirical regimes: fixed-model lift, frozen transfer within a model family, adapted transfer across providers, and held-out task generalization. The transferable object becomes more abstract as distance grows. Nearby models can share an exact control profile. Across providers, the runtime, runbook structure, applicable practices, and development principles remain reusable, although the profile requires adaptation. Across task distributions, transfer resides in the method for locating and protecting consequential execution boundaries. The question is *what transfers as model and task distance increase*.

4.1 Evaluation Regimes and Evidence Boundaries

Terminal-Bench 2.1. Terminal-Bench 2.1 (Merrill et al., 2026) contains 89 tasks. We run five trials per task, for 445 trials in a full evaluation. The primary metric is trial-level success rate. We also report *five-trial task*

coverage, the number of tasks solved at least once across their five trials. This quantity is sometimes reported as Pass@5, but it measures empirical coverage over five observed trials rather than single-run reliability.

Our Terminal-Bench experiments use Codex-based agents with GPT-5.5 xhigh, GPT-5.6 Sol xhigh, GPT-5.6 Luna, and DeepSeek-V4-Flash. Rows labeled *reference* reproduce the corresponding published or public-submission result; rows labeled *our run* report our evaluation. The GPT-5.6 Sol xhigh submission uses **statem-Codex** agent version 0.144.1. Its public submission record contains 445 trials, 439 no-error completions, six **AgentTimeoutError** trials, 1.178 billion tokens, a standard error of 0.87 percentage points, and a submission-reported model cost of \$1,062.95. It passes all ten automated configuration and static checks in the submission pipeline.

Development, freezing, and adaptation. We first develop the Terminal-Bench control profile with GPT-5.5. For the GPT-5.6 Sol and Luna evaluations, the versioned profile and its activation policy are frozen: no prompt, state, routing, check catalog, default, or repair policy is revised using target-model evaluation outcomes. Any permitted run-local check is instantiated only from task-visible information under the frozen activation policy, is recorded for that execution, and is not promoted into later trials. The DeepSeek experiment first measures this same frozen profile, then explicitly permits a new adaptation phase before the final evaluation.

Golden rules are human-specified constraints on profile development: prefer minimal reusable control, route from visible task semantics rather than task identity, and separate development feedback from frozen evaluation. They govern profile evolution and are distinct from the checks executed inside a run.

In every setting, profile construction may use the visible task specification, workspace artifacts, public documentation, and observable execution feedback. It does not use hidden tests, verifier implementations, public solutions, answer artifacts, task hashes, or manually enumerated task identifiers as routing keys. This boundary prevents direct answer encoding while allowing failure-driven improvement of the harness.

BusinessBench. BusinessBench (Yang et al., 2026) tests generalization at the task-family level. We construct one profile per treated family, refine it on development instances, and freeze it before the first held-out evaluation. Later refinements using aggregate evaluation feedback are reported separately as post-evaluation diagnostic validation. Each instance has one stochastic trajectory per arm, so family-macro and instance-micro averages answer complementary questions.

Cost accounting. For DeepSeek, *final-evaluation expenditure* is the realized API charge for producing the reported final-score evidence. *Adaptation expenditure* includes all recorded API charges from the provider-specific profile-development iterations. We report both components and their sum. Public comparator rows list submission-reported model cost separately from the realized API expenditure of our runs. The \$1,062.95 attached to the GPT-5.6 StateM submission is the pipeline’s API-equivalent model-cost estimate, not the authors’ realized out-of-pocket expenditure under a Codex Pro plan.

4.2 Same Model, Better Harness: A Model-Generation-Sized Gain

Holding model weights fixed isolates the contribution of the harness. With GPT-5.5 xhigh, our StateM system records 92.1% on Terminal-Bench 2.1, compared with the 83.1% GPT-5.5 Codex reference. This nine-point difference closes a model-generation-sized gap without changing the underlying model. Across five trials, GPT-5.5 with StateM solves 88 of the 89 tasks at least once, corresponding to 98.9% five-trial task coverage. Its 92.1% mean is also numerically above the separately reported 91.9% GPT-5.6 Sol ultra reference in Fig. 3.³

The gain persists with the stronger model. GPT-5.6 Sol xhigh with StateM records

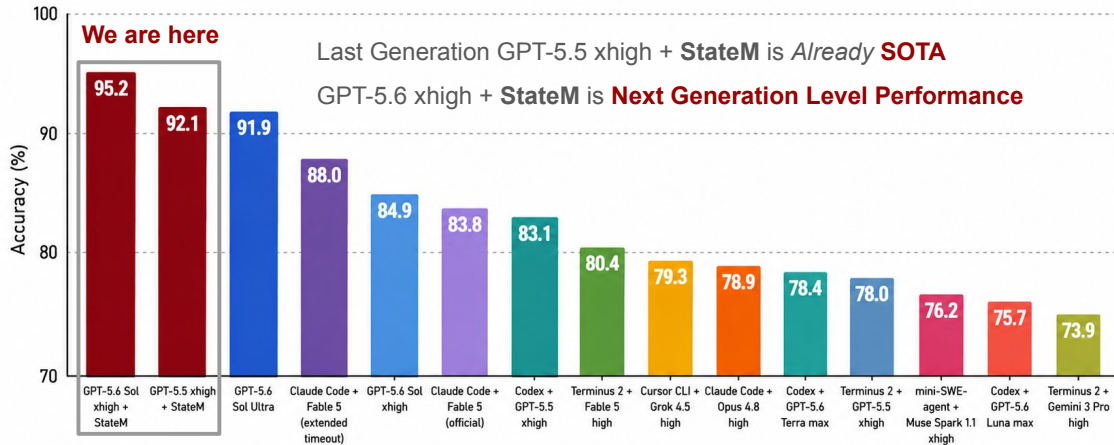
$$\frac{424}{445} = 95.28\%$$

raw accuracy, reported as 95.3% at one decimal, compared with the 84.9% GPT-5.6 Sol xhigh reference. This is a 10.38-point difference, or 10.4 points at one decimal. The system solves every one of the 89 tasks at least

³The public GPT-5.5 reference is 83.15% over 445 trials with Codex agent version 0.125.0 and \$2,059.19 reported model cost; we round its score to 83.1%. It is a model-matched public reference rather than a newly rerun, agent-version-matched A/B control. The 91.9% Sol ultra value is likewise a numeric frontier reference from a different high-compute configuration

Terminal-Bench 2.1 Leaderboard

Accuracy (%) on Terminal-Bench 2.1. Includes official verified leaderboard entries from tbench.ai, plus reported results for GPT-5.6 Sol xhigh + StateM, GPT-5.5 xhigh + StateM, GPT-5.6 Sol Ultra, and Claude Fable 5.



Source: official entries from tbench.ai Terminal-Bench 2.1 leaderboard; additional reported results added by author.

Figure 3 A model-generation-sized harness gain on Terminal-Bench 2.1 GPT-5.5 xhigh with StateM records 92.1%, compared with the 83.1% GPT-5.5 reference. GPT-5.6 Sol xhigh with the frozen StateM profile records 95.28% raw accuracy, compared with the 84.9% Sol xhigh reference. Reference configurations and our runs are visually distinguished; 95.28% is shown as 95.3% at one-decimal precision

once across five trials. Submission status and adjudication sensitivity are disclosed once in the Introduction; throughout this section, 95.28% denotes the raw public-submission result.

Under the reference harness, moving from GPT-5.5 to GPT-5.6 Sol changes the score from 83.1% to 84.9%, a 1.8-point model-generation shift. The StateM reference differences are +9.0 points on GPT-5.5 and +10.4 points on GPT-5.6 Sol. Here, the execution harness changes completed-task performance far more than the observed generation-to-generation model shift. Model scaling expands what the agent can do; harness scaling determines how reliably that capability becomes a completed job.

4.3 Transfer Follows Model Distance

The fixed-model gains survive model changes at different levels of abstraction, producing a hierarchy rather than a binary transfer outcome.

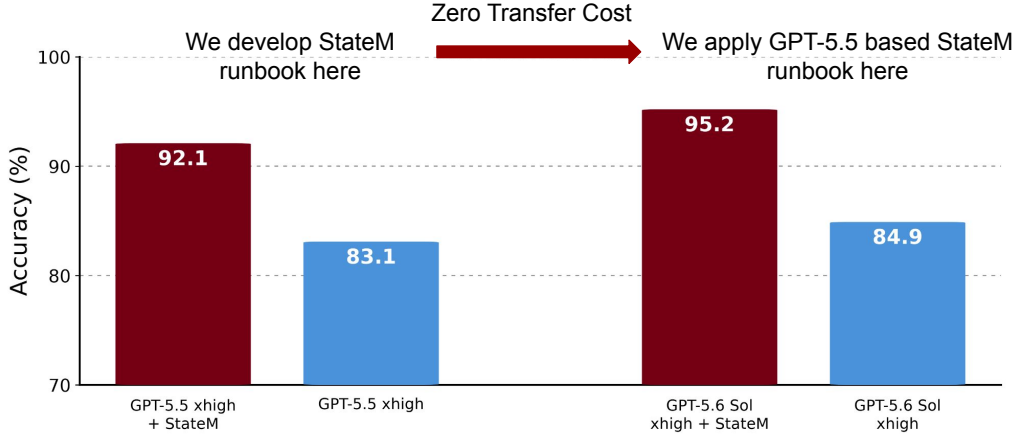
Frozen transfer within the GPT family. The Terminal-Bench profile is developed with GPT-5.5 and then frozen before application to GPT-5.6. With no GPT-5.6-specific change to its control logic, the profile accompanies the 84.9% to 95.28% Sol xhigh difference reported above and raises GPT-5.6 Luna from 76.7% to 85.4%, a gain of 8.7 points. Luna with StateM therefore numerically exceeds the 84.9% Sol xhigh reference despite using the lower-cost model tier.

Closely related GPT generations and variants share enough recurring execution failures for the same executable control profile to remain useful. The transferred object is concrete: the states, activation policy, routing, checks, and repair structure remain unchanged. “Frozen transfer” means zero target-model runbook changes, not zero evaluation cost.

The provider boundary changes the transferable object. Directly applying the frozen GPT-developed profile to DeepSeek-V4-Flash changes the full-suite score from 82.7% to 82.0%. Exact-profile transfer fails across this provider boundary. Because StateM preserves substantial autonomy in the base agent, provider-specific behavior remains consequential even when the task set and control interface are held fixed.

Cross-version StateM Compatibility on Terminal-Bench 2.1

Accuracy (%) for four configurations showing that the same harness improves both GPT-5.6 Sol xhigh and GPT-5.5 xhigh.



Source: official entries from tbench.ai Terminal-Bench 2.1 leaderboard and OpenAI Tech Report; additional reported StateM results added by author.

Figure 4 Frozen cross-generation transfer with zero target-model runbook changes The profile developed with GPT-5.5 transfers unchanged to GPT-5.6 Sol xhigh, where the reference difference grows from +9.0 to +10.4 points. The evaluated profile is frozen before any GPT-5.6 outcomes are observed

Cross-provider adaptation does not begin from scratch. The generic runtime, high-level runbook structure, routing strategy, already applicable controls, failure-analysis loop, and golden rules remain available. Adapting the concrete practices to DeepSeek’s residual failure distribution produces the cost frontier reported in Sec. 4.4. Transfer follows distance: exact profiles transfer across nearby GPT models, while development principles and the control structure transfer across providers.

4.4 A \$15 Frontier Run: Moving the Cost–Accuracy Curve

DeepSeek-V4-Flash marks the point where exact-profile transfer fails but profile adaptation succeeds. Its 82.7% baseline changes to 82.0% under direct frozen transfer of the GPT profile. Starting from the same runtime, runbook structure, applicable controls, and golden rules, provider-specific adaptation raises the full 89-task standard-timeout result to

$$\frac{392}{445} = 88.09\%,$$

a gain of 5.39 points over the 82.7% full-suite baseline. On the disclosed latency-stable common core, which excludes only `gpt2-codegolf`, the same 392 successes give

$$\frac{392}{440} = 89.09\%.$$

For this configuration, the remaining task is limited by the standard timeout rather than absolute solvability: `gpt2-codegolf` is 0/5 under the standard timeout, but DeepSeek-V4-Flash with StateM solves it in 3 of 5 trials under an extended per-task timeout. Combining those disclosed trials with the 88-task common-core result yields a descriptive full-suite aggregate of

$$\frac{392 + 3}{445} = \frac{395}{445} = 88.76\%,$$

which rounds to 88.8% and matches the separately reported GPT-5.6 Sol max score at one-decimal precision.⁴

⁴The standard-timeout full-suite result is $392/445 = 88.09\%$; the common-core result is $392/440 = 89.09\%$; and the descriptive aggregate is $395/445 = 88.76\%$ after evaluating only `gpt2-codegolf` with the extended timeout. Reporting all three numbers keeps the operating condition of each result explicit

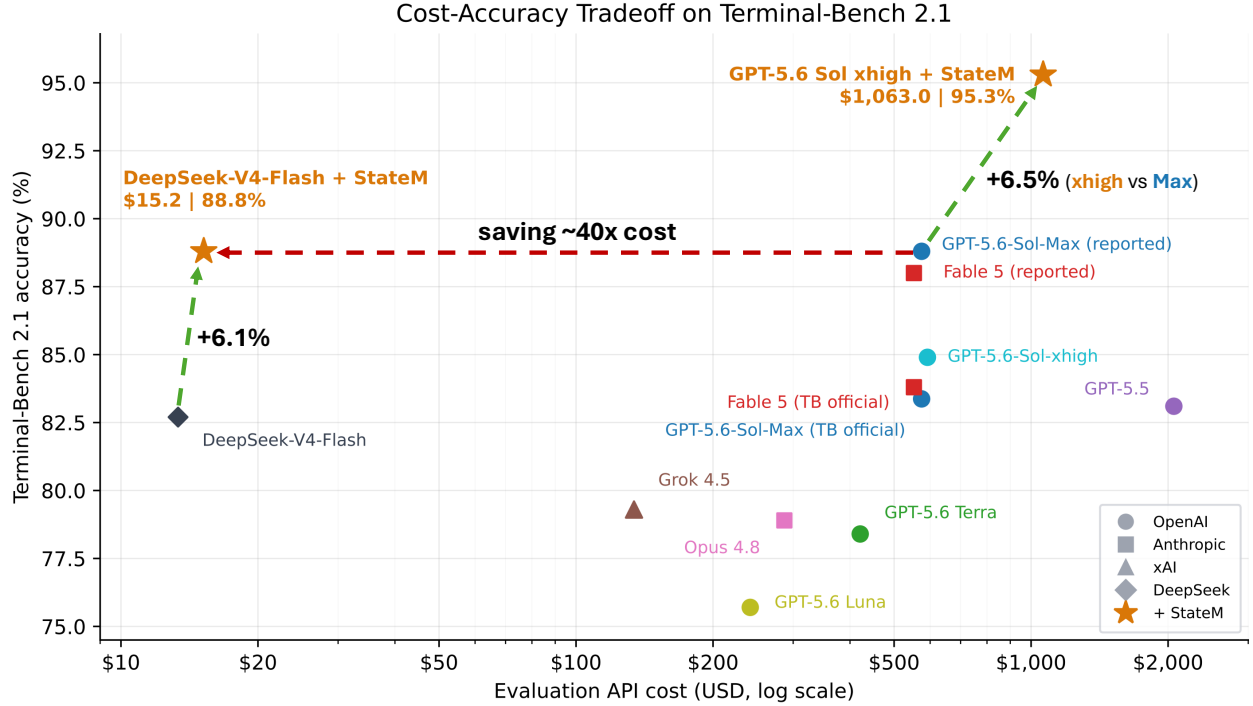


Figure 5 StateM moves both the quality and cost frontiers on Terminal-Bench 2.1 The orange points are matched score–cost pairs from our runs: GPT-5.6 Sol xhigh + StateM at 95.28% raw and \$1,062.95 submission-reported model cost, and DeepSeek-V4-Flash + StateM at 88.76% descriptive accuracy and \$15.20 realized final-evaluation expenditure. The horizontal 88.8% reference denotes the separately reported GPT-5.6 Sol max score. The public \$574.68 GPT-5.6 Sol max submission is plotted at its matched 83.37% raw score. The upper-left direction is better

The cost frontier moves even more sharply. The complete DeepSeek final-score evidence costs \$15.20 in realized API charges. Provider-specific adaptation costs \$37.02, bringing all recorded DeepSeek adaptation and final-evaluation expenditure to \$52.22. In contrast, the public GPT-5.6 Sol max Codex submission records \$574.68 in model cost. The DeepSeek evidence uses 2.65% of that submission’s recorded cost, or 1/37.8 as much; its standard-timeout full-suite score is 88.09%, compared with 83.37% raw for the matched public GPT artifact. Even the entire \$52.22 adaptation-and-evaluation campaign uses about 1/11 of that amount. The separate OpenAI-reported Sol max score is 88.8%, which the descriptive DeepSeek aggregate matches to one decimal.⁵

Harness transfer changes the economics of deployment. Exact controls did not survive the provider boundary unchanged, but the reusable runbook structure and development principles made adaptation cheap enough to move a lower-cost model sharply upward on the quality–cost frontier. The choice is no longer simply “buy the strongest model”: one can invest in a harness that turns a cheaper model into a stronger system.

4.5 Task Generalization Follows the Control Boundary

Terminal-Bench tests one shared profile across heterogeneous tasks. A coherent task family, by contrast, has its own tools, invariants, and completion boundary, and a profile designed for another family may impose irrelevant procedure. We evaluate StateM on BusinessBench (Yang et al., 2026) using Codex with GPT-5.6 Luna. Family-level profiles are developed on one split and tested on held-out instances from the same family; together, the families test whether the development method remains useful across heterogeneous work.

⁵The 88.8% Sol max score and the \$574.68 public-submission cost originate from different evaluations. The \$574.68 submission reports 83.37% raw before adjudication and 76.18% after adjudication. We therefore use 88.8% as a score-only reference and plot \$574.68 only at its matched submission score

Scope / task family	Codex CLI	StateM–Codex	Δ
<i>Frozen one-shot aggregate results</i>			
Held-out, family macro	84.67	85.22	+0.55
Held-out, instance micro	84.44	85.78	+1.34
Development	86.07	91.71	+5.64
All StateM-treated	84.76	88.72	+3.96
<i>Exploratory mechanism-matched held-out subgroup</i>			
Budget Approval + Machine Operating, family macro	71.91	81.94	+10.04 [†]
<i>Frozen Round-1 family aggregates</i>			
budget-approval	62.91	75.12	+12.21
machine-operating	90.79	100.00	+9.21
refactorbench	80.56	77.78	−2.78
webarena	88.00	92.00	+4.00
webtest	98.25	98.75	+0.50
woocomerce-stock	92.59	88.89	−3.70
<i>Abstention control, excluded from StateM aggregates</i>			
attendance-payroll	88.43	abstained	n/a

Table 1 Frozen one-shot BusinessBench results with Codex + GPT-5.6 Luna The first four rows summarize the six treated families; family rows report the corresponding Round-1 aggregates. Attendance is excluded because no StateM workflow is applied. [†]The 10.04-point subgroup delta is computed from unrounded source values; the displayed values differ by 10.03 after independent rounding. The 80.56 RefactorBench Round-1 baseline and the later 80.56 StateM development score in Table 2 belong to different runs and splits

The benchmark contains 477 eligible instances across seven families. The protocol abstains from applying a StateM workflow to **attendance-payroll**; all 72 treatment-arm runs in that family receive no StateM intervention and are excluded from efficacy aggregates. The treated set therefore contains 405 instances across six families, or 810 genuine treatment/control executions. GPT-5.6 Luna task agents execute development instances and propose reusable changes from their failure traces; a GPT-5.6 Sol xhigh hyper-agent evaluates family-level generality and compatibility before reconciling the accepted changes. The resulting profile is frozen before the first held-out evaluation.

Frozen transfer improves held-out performance. On the untouched one-shot held-out split, the equal-family macro average improves from 84.67% to 85.22%, a gain of 0.55 points. The instance-weighted micro average improves from 84.44% to 85.78%, a gain of 1.34 points. Development performance improves from 86.07% to 91.71% (+5.64), and performance over all treated instances improves from 84.76% to 88.72% (+3.96).

The largest gains are concentrated in two structurally matched families. Budget Approval improves from 62.91% to 75.12% (+12.21), and Machine Operating improves from 90.79% to 100.00% (+9.21). Their equal-family held-out macro average rises from 71.91% to 81.94%, a 10.04-point gain computed from the unrounded family values. Both families require the workflow to preserve explicit constraints, intermediate state, and completion conditions across standardized tools. Budget Approval benefits from exact-decimal calculation, policy reconciliation, and mandatory-effect closure. Machine Operating benefits from task-derived query planning, data-plane execution, interval coverage, and durable publication.

Negative transfer identifies the wrong boundary. The first frozen profiles do not improve every family. RefactorBench changes from 80.56% to 77.78% (−2.78), and WooCommerce Stock changes from 92.59% to 88.89% (−3.70). In both cases, the problem is not simply too little control. The control is attached to the wrong execution boundary.

RefactorBench initially overemphasizes minimality and backward compatibility but fails to close explicit code-migration obligations. A later, thinner profile tracks visible obligations, verifies required signature and

Post-evaluation validation scope	Codex / prior arm	Refined StateM	Interpretation
Held-out, instance micro	n/a	86.67	Selectively refined aggregate
Development	n/a	92.21	Selectively refined aggregate
All StateM-treated	n/a	89.42	Selectively refined aggregate
refactorbench , overall	76.39	79.17	Matched rerun
refactorbench , development	75.00	80.56	Obligation-and-closure profile
refactorbench , reused held-out	77.78	77.78	No held-out change
webtest , overall	99.75	99.75	Saturated
webtest , development	100.0	100.0	Saturated
webtest , reused held-out	99.5	99.5	Saturated
woocomerce-stock , overall	86.42	90.12	Invariant-matched profile
woocomerce-stock , reused held-out	85.37	90.24	Invariant-matched profile

Table 2 Post-evaluation diagnostic validation after selective profile refinement These results use aggregate evaluation feedback and are separated from the untouched one-shot evidence in Table 1. The first three rows are refined StateM aggregate scores; the remaining rows are matched reruns or split-level diagnostics

call-shape changes, and performs one repository-wide stale-reference closure. In a post-evaluation matched rerun, the overall score changes from 76.39% to 79.17%. On the development split, the revised StateM arm improves from 75.00% to 80.56%; on the reused held-out split, it remains 77.78%. The rerun locates the original negative transfer in profile mis-specification: the useful intervention is less general compatibility procedure and more direct obligation closure.

WooCommerce exhibits a different boundary error. Its first profile contains substantial procedure but fails to preserve the cross-system invariants that govern inventory entities, destination-specific deduplication, irreversible email actions, and receipts. After replacing generic control with these invariants, a post-evaluation matched rerun improves overall performance from 86.42% to 90.12%, including an improvement from 85.37% to 90.24% on the reused held-out split.

WebArena is dominated by retrieval and analysis. Query modifiers, page state, sorting, and pagination matter more than a mutation-oriented workflow. Its frozen evaluation contains only three non-tied A/B outcomes across 25 instances: development has 2 wins, 10 ties, and 0 losses; held-out has 0 wins, 12 ties, and 1 loss. A proposed minimal query-state profile has not yet been evaluated due to model capacity constraint. WebTest is nearly saturated: later matched reruns reach 100.0% on development and 99.5% on the reused held-out split for both arms, or 99.75% overall.

Across the seven families, **harness generalization follows mechanism match, not task diversity**. RefactorBench requires code-contract and stale-reference closure; WebArena requires query and page-state control; WebTest primarily requires a tool-execution boundary; WooCommerce requires entity, destination, and irreversible-side-effect invariants; and Attendance shows that abstention can be the correct control decision. A strong profile is not a maximal universal workflow. It binds the minimum invariants likely to drift and verifies them where a violation becomes consequential. Appendix Table 5 lists the generalized controls distilled from these families.

4.6 Failure-Boundary Evidence: Where StateM Intervenes

Control-signal dilution and mutable-state ambiguity create the operational pressures that make long runs difficult. Epistemic, procedural-compliance, and procedural-memory gaps locate the resulting failures. StateM supplies a distinct control point for each gap.

For an *epistemic gap*, a state-local `in_hook` can reactivate relevant domain knowledge, tool guidance, or task context at the phase where it becomes useful. For a *procedural-compliance gap*, states, checks, transition guards, and evidence requirements can block a configured incomplete handoff and keep the run in a repairable state. For a *procedural-memory gap*, a selected lesson from one execution can become a versioned prompt, check, activation rule, or recovery path for later independent runs.

Task	Codex CLI	StateM-Codex	Δ	Associated StateM control
<code>configure-git-webserver</code>	0/5	5/5	+5	Service/deploy state and consumer-facing verification
<code>dna-insert</code>	0/5	5/5	+5	State-local biological and primer contract checks
<code>dna-assembly</code>	1/5	5/5	+4	Transition gate for primer, Tm, and assembly invariants
<code>filter-js-from-html</code>	0/5	4/5	+4	HTML/script extraction boundary checks and negative controls
<code>db-wal-recovery</code>	2/5	5/5	+3	Preflight preservation before destructive consumers
<code>sanitize-git-repo</code>	4/5	5/5	+1	Modification-range verification
<code>protein-assembly</code>	2/5	5/5	+3	Constraint checklist and self-review
<code>pypi-server</code>	3/5	5/5	+2	Service-lifetime and readiness gate
<code>install-windows-3.11</code>	3/5	5/5	+2	VM setup and lifecycle completion
<code>pytorch-model-recovery</code>	3/5	5/5	+2	Task-visible model evidence
<code>qemu-alpine-ssh</code>	2/5	4/5	+2	Service readiness and repeated consumer check
<code>extract-moves-from-video</code>	0/5	2/5	+2	Candidate-first bounded refinement

Table 3 Representative Terminal-Bench 2.1 task-level improvements for GPT-5.5 Each result is the number of successful trials out of five for one task. The listed controls were proposed and implemented by GPT-5.5 Codex under the human-specified high-level design and golden rules. They are selected from visible task semantics and workspace evidence, without using task identifiers, task names, hashes, hidden tests, verifier implementations, or answer artifacts as activation keys

Table 3 reports representative GPT-5.5 task-level evidence. The largest gains cluster around consequential boundaries: service readiness before handoff, contract satisfaction before committing a transformation, preservation before destructive operations, and evidence closure before completion. The table associates each gain with the StateM control active in that task; the association is not a component ablation.

The `configure-git-webserver` task isolates this distinction. The baseline agent can configure Git, SSH, hooks, and an HTTP server, yet scores 0/5 because it does not reliably preserve and validate the required end-to-end live state. With StateM, final handoff is gated on fresh consumer-facing evidence: the agent must materialize a clone-commit-push-curl path before leaving verification. If verification perturbs the environment, the run remains in a repairable state until final-state consistency is restored. StateM adds no new component-level capability in this case. It composes, checks, and closes capabilities already available to the model before the consequential handoff boundary. The result moves from 0/5 to 5/5.

Epistemic hooks determine which knowledge is active at a state; checked transitions determine whether a known procedure is completed; and versioned practices determine whether a lesson survives into future runs. StateM preserves the base agent’s freedom inside each phase while adding the minimum persistent control where a recoverable error would otherwise become final.

4.7 Harness Learning as Selective Procedural Memory

The runbook is both a runtime control surface and a selective memory for lessons that persist across independent executions. We call a reusable experience-derived intervention a *practice*. A practice may be a state-local instruction, a check, a constraint, an activation condition, or a verification action triggered when visible evidence indicates elevated downstream failure risk. Encoding the lesson in a versioned profile makes it inspectable and executable; a later run no longer has to rediscover it from a postmortem paragraph.

From one failure to one candidate practice. At the single-task level, learning follows a concrete loop: execute the current profile, observe failure or verifier feedback, attribute a candidate cause, abstract a reusable

lesson, encode a profile change, and validate it again. Because the runbook is agent-native, the executor and maintainer can be the same agent. The agent can change a state-local prompt, add an evidence check, revise a transition, or introduce a repair path without reimplementing the entire reasoning process as a collection of controller-owned graph nodes.

From task traces to a family profile. Multi-task settings make the abstraction step decisive. A lesson from one development instance should not automatically become a rule for the family. In BusinessBench, GPT-5.6 Luna task agents execute development instances and propose reusable changes. A stronger hyper-agent evaluates their family-level generality and compatibility before reconciling them into a shared family profile. The frozen one-shot evaluation tests whether that abstraction survives on held-out tasks.

One broad benchmark requires sparse routing. Terminal-Bench requires one shared profile to cover 89 heterogeneous tasks, while the accumulated trajectories exceed what a hyper-agent can review in one context. Repairing tasks one at a time causes control drift: a local rule can add unnecessary procedure or damage a different task. We use a thin shared runbook, generalizable routing from visible task semantics, and late state-local checks activated only when the current task and workspace evidence make them relevant. For epistemic gaps such as statistical reasoning in `raman-fitting`, the corresponding `in_hook` can expose general public domain knowledge gathered by the hyper-agent. Human input supplies the high-level architecture and golden rules; GPT-5.5 Codex produces most of the Terminal-Bench harness implementation and performs much of the low-level trace analysis and proposal generation.

The BusinessBench iterations show why this memory must be selective. More remembered procedure is not better memory. RefactorBench improves when repeated compatibility review is replaced by a smaller obligation-and-closure practice. WooCommerce improves when generic procedure is replaced by the actual cross-system invariants. WebArena appears to require a still lighter query-state boundary. Attendance shows that the correct policy can be not to introduce a StateM workflow at all. Effective procedural memory requires addition, revision, and removal.

What the harness should not remember. Failure-driven optimization can preserve the wrong lesson. The Terminal-Bench video tasks do not fully specify target-frame precision. During development, the hyper-agent introduces default precision values after observing benchmark behavior. These defaults may be operationally useful, but they also show how an ambiguous specification can be resolved post hoc and then stored as if it were a universal practice.

Evaluator feedback enters memory through the same route. In the DNA insertion tasks, the verifier selects the left-most valid insertion boundary even though that convention is absent from the visible task description. Repeated feedback can lead the profile to reproduce that convention without ever reading verifier code. The resulting behavior agrees with the evaluator, but its semantics come from the evaluator rather than the stated task contract.

The first BusinessBench profiles reveal a third failure mode: even a valid failure can be abstracted incorrectly. RefactorBench receives excessive compatibility procedure, retrieval-oriented WebArena receives a workflow that is too heavy, and early WooCommerce control misses the invariants that connect its systems. Stronger-agent or human review matters at this level. Failure traces provide experience; deciding which abstraction deserves to become durable control is itself a difficult reasoning task.

The design rule is direct: **experience must be filtered before it becomes memory**. Harness scaling is an abstraction problem, not rule accumulation. The objective is to remember the consequential boundary, not every failure trace.⁶

4.8 Operational Endurance: A 22-Hour Development Run

StateM was built for an operational requirement beyond benchmark accuracy: a long-running agent must preserve its place and obligations as the interaction grows. In one Terminal-Bench profile-development run, the hyper-agent continued for 22 hours across long interaction history, context refresh or compaction, and

⁶BusinessBench family sizes are unequal and each instance has one stochastic trajectory per arm. Only the first frozen held-out evaluation is untouched; later results are explicitly post-evaluation validation. StateM can organize available knowledge and execution control, but it cannot remove limits imposed by the base model, tools, environment, or execution budget

stop-hook continuation. Throughout the run, the durable StateM record provided the current phase, transition history, unresolved obligations, and recovery anchor. The observed slowdown came from accumulated terminal output that remained unfolded in the interface, rather than loss of the StateM control state.

The runtime sustains a day-scale development loop while keeping procedural state external to the changing model context; the result does not imply unbounded execution. Together with the benchmark results, this completes the transfer hierarchy: exact runbooks transfer across nearby models, development principles transfer across providers, and control-boundary abstractions transfer across tasks.

5 Discussion: Runtime Design for Harness Scaling

Harness scaling requires a runtime that is both agent-native and enforceable. The experiments above highlight a practical requirement that is easy to miss when viewing a harness only as a fixed workflow: the runtime itself must be cheap to modify. Conventional state-machine and graph-based agent runtimes provide explicit orchestration and strong control, but typically externalize substantial reasoning into manually constructed nodes, edges, and handlers. As the control policy evolves, modifying this structure becomes an engineering task of its own, making rapid failure-driven iteration increasingly heavy. At the other extreme, agent-native systems such as Codex and Claude Code allow the model to plan and revise its behavior fluidly inside the agent loop, but much of this control remains model-mediated: plans, checklists, and self-review can guide behavior without providing the same persistent, auditable, and externally enforceable transition semantics. STATEM is designed to combine these two properties. The agent continues to reason and act in its native interaction loop, while states, hooks, transition conditions, and practices provide a control surface that both agents can modify and humans can inspect and enforce. This makes the control layer itself amenable to rapid iteration—a prerequisite for *harness scaling*, where repeated execution experience is used to continuously improve the runtime rather than only the model or prompt.

Beyond single-agent control. This work focuses on controlling the execution of a single autonomous agent, but the same separation of state, context, and permissions also motivates multi-agent runtimes. In long-running work, a single agent can become overconfident in an early task interpretation and continue reasoning inside a context shaped by its own previous decisions. Separating agents by role can provide different context views and independent decision boundaries, while role-specific read/write permissions can restrict which artifacts, state, or actions each agent may modify. Such isolation may be useful for review, verification, delegation, and recovery from decisions that would otherwise become self-reinforcing within one context. We are exploring these extensions, but they introduce a separate set of coordination, state-sharing, and permission-design questions. We therefore restrict the present work to single-agent control flows and leave role-isolated multi-agent runtimes to future work.

6 Conclusion

This work studies harness scaling as a capability axis complementary to model scaling. We introduce STATEM, an agent-native control layer that externalizes mutable execution state, refreshes phase-local context, and checks consequential transitions while preserving the unified reasoning loop of a general-purpose CLI agent. The agent-native and enforceable features of STATEM make harness scaling feasible in efficiency and cost.

The experiments expose two different frontiers. At the quality frontier, GPT-5.6 Sol xhigh with a runbook frozen from GPT-5.5 records 95.3% raw accuracy on Terminal-Bench 2.1, or 424 successful trials out of 445, with at least one success on every task. This remains a public, pre-adjudication submission rather than a finalized leaderboard result. At the cost frontier, an adapted DeepSeek-V4 Flash system reaches 88.09% on the full benchmark under standard timeouts and 89.09% on the disclosed 88-task common core. Its final evidence costs approximately \$15 in DeepSeek API usage, while the complete recorded DeepSeek API expenditure for adaptation and evaluation is \$52.22.

The transfer results show that procedural control is hierarchical rather than universal. Concrete practices transfer unchanged across the GPT-5.5 and GPT-5.6 generations, but do not transfer unchanged to DeepSeek-V4 Flash. Across providers, the runtime and failure-driven development methodology remain reusable, while model-dependent practices must be adapted. BusinessBench yields the corresponding task-side lesson: frozen

aggregate generalization is positive but modest, and the largest gains occur where control is sparse and matched to an explicit execution boundary.

More broadly, our results distinguish three ways in which an otherwise capable model can fail. Relevant knowledge may be unavailable at the decision point; a lesson from prior experience may not be retained or activated; or an active procedure may not be completed. STATEM addresses these operational gaps through state-local context, versioned procedural practices, and checked transitions, respectively.

STATEM does not make every model universally stronger, and one runbook does not fit every workflow. Its value is to make execution failures inspectable and to convert selected lessons into reusable control. The central scaling problem is therefore not how many rules a harness remembers, but which lessons should persist, where they should intervene, and whether they survive a change of model or task. Model scaling expands what an agent can do; harness scaling helps the system remember what experience has taught it to do without training, and ensures that it actually does so before proceeding.

7 Acknowledge

We thank Zekai Li and Mengxuan Wu for discussions and feedback on this work.

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A Runbook example

We here share an example of coding agent runbook:

```
name: coding-agent-loop
initial: start

nodes:
  start:
    prompt: |
      Initialize the run context from durable project files. Read the task description,
      progress.md, StateM history, architecture notes, and any project rules
      before planning. It is fine to inspect the full runbook with
      'StateM state'; use 'cur', 'next', and 'goto' to execute the current step
      with discipline. Do not use '/clear' as part of the normal loop; use the
      session_refresh node and safe compaction when a later cycle needs cleaner
      context.
    in_hook:
      type: message
      text: "Load the task brief, progress.md, StateM history, and any
      architecture notes before planning."
    before_transfer:
      type: checklist
      items:
        - Task brief was read
        - Relevant files were inspected
        - Constraints were noted

  plan:
    prompt: |
      Generate an implementation plan from the task context, current progress,
      architecture notes, and project rules.
    before_transfer:
      - type: manual
        prompt: "Plan has been reviewed and is concrete enough to execute."
      - type: predicate
        path: ../progress.md
        exists: true
        non_empty: true

  execute:
    prompt: |
      Execute the plan while keeping edits scoped. Update progress before
      leaving this state. If the concrete task or implementation approach
      creates state-specific review needs, register current-entry dynamic
      checks before leaving execute.
    in_hook:
      type: message
```

```

    text: "Use 'StateM dynamic path' and 'StateM dynamic write' to register
    task-specific current-entry checks when the implementation needs them."
before_transfer:
  type: checklist
  items:
    - Implementation matches the plan
    - Tests or verification were run
    - No unrelated files were changed
out_hook:
  type: message
  text: "Update progress.md before moving on."

review:
  prompt: |
    Review the implementation before handing work back to the user. Compare
    the code against the plan, project docs, constraints, and golden rules.
    If the implementation differs from the plan, decide whether the change is
    justified and make sure the new consideration is recorded in progress.md
    or the final handoff. Prefer returning to execute when the review finds a
    fixable gap.
  before_transfer:
    type: checklist
    items:
      - Implementation follows the task and current plan, or deviations are justified
      - Relevant docs, architecture notes, and local conventions were followed
      - Constraints and golden rules were checked explicitly
      - Edited file scope is appropriate and unrelated changes were not reverted
      - Tests, lint, or equivalent verification were run, or blockers are recorded
      - Remaining risks and follow-ups are clear enough for the user handoff
  out_hook:
    type: message
    text: "Record review notes, verification results, and any remaining risks
    before leaving review."

handoff:
  prompt: |
    Hand the run back to the user. Summarize what changed, verification run,
    current state, known risks, and the recommended next command if work will
    continue later.
  in_hook:
    type: message
    text: "Before replying to the user, read progress.md and StateM history
    so the handoff is grounded in durable state."

session_refresh:
  prompt: |
    Use this node only after a complete review pass when another loop will
    continue and the context contains stale attempts or noisy intermediate
    output. Generate a safe compaction instruction with
    'StateM compact-prompt --run-id <id>', compact through the host UI, then
    restore attention with 'StateM cur' and 'StateM history'.
  before_transfer:
    type: checklist

```

```

    items:
      - progress.md and StateM history contain the durable facts needed after compaction
      - Safe compaction was applied, or the context was judged clean enough to skip it
      - Current state and next transition were rechecked after compaction

edges:
  - from: start
    to: plan
    condition: "Initial context is loaded."

  - from: plan
    to: execute
    condition: "Plan is ready for execution."

  - from: execute
    to: review
    condition: "Implementation is ready for review."

  - from: review
    to: execute
    condition: "Review found changes that need another execution pass."

  - from: review
    to: handoff
    condition: "Review passed, or remaining issues and blockers are documented for the user."

  - from: review
    to: session_refresh
    condition: "Review passed and another loop will continue in a cleaner session."

  - from: session_refresh
    to: plan
    condition: "Session context is compacted or intentionally kept, and state was restored."

```

B Our Experiment Equipment and Budget

The main computation resource, is personal Codex pro plan and Macbook Pro 2025 (M4 chip). Thanks again to Codex's pro plan so we can have chance to do such an individual research work within 200\$ budget, and the actual usage is less than 125\$ of it. For the reported formal run submission, we use AWS m7i.4xlarge, because daytona would encounter sandbox/verifier/network/pytorch-install timeout more frequently, and personal Macbook Pro 2025 (M4 chip) is not compatible with tune-mjcf kernel requirement. On a personal Macbook Pro 2025, GPT-5.5 xhigh can reaches around 91% pass rate on Terminal Bench 2.1.

C Additional Figures and Tables

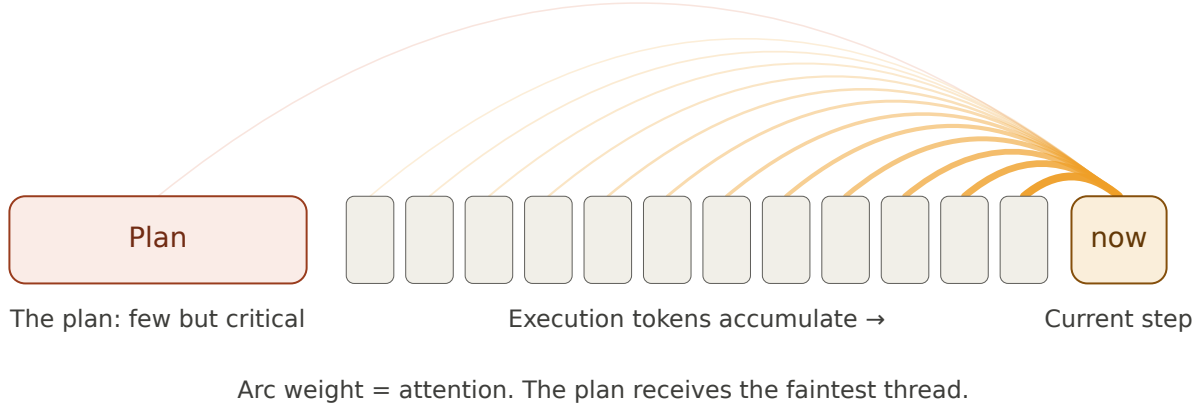
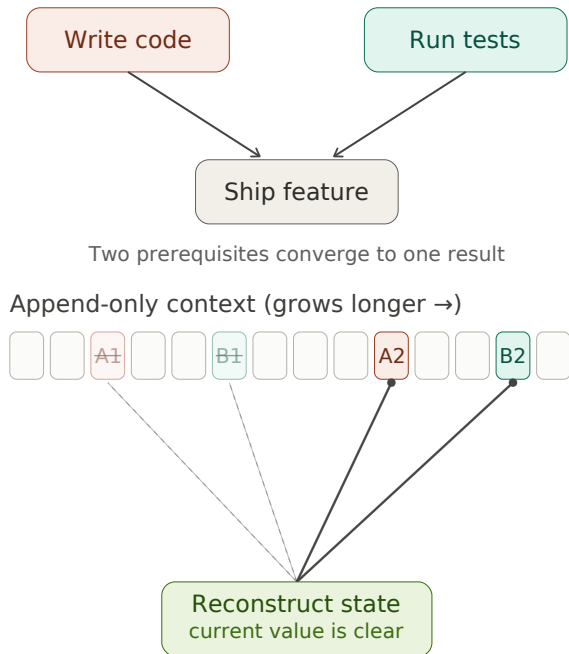


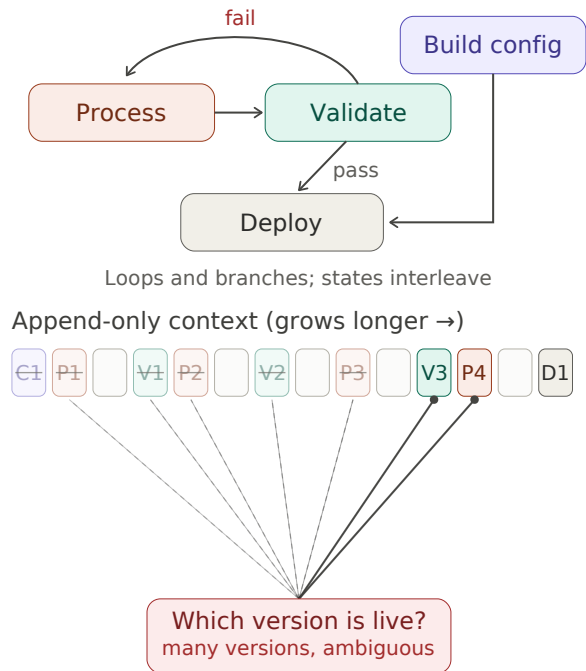
Figure 6 Attention flow for mixed plan and execution tokens. Plan tokens are at higher abstraction level, which is much fewer but critical. They are diluted by execution tokens in a long run.

Simple task (linear dependency)



(a) Simple task with linear dependencies: few state updates are appended, so the latest version of each item is easy to locate and the current state recovered without ambiguity.

Complex task (loop + branch)



(b) Task with loops and branches: repeated iterations append many interleaved versions of the same item, making the current state ambiguous.

Figure 7 State maintenance degrades with task complexity. Under an append-only context with no in-place editing, each update is appended rather than written back. For a simple task (left) the latest value is still recoverable; for a task with loops and branches (right) the interleaved versions make the current state hard to reconstruct.

System			Result source / profile status	Evaluation scope	Score / five-trial coverage
GPT-5.5 Codex	xhigh	+	Published reference	89 tasks, 445 trials	83.1%; coverage not reported
GPT-5.5 StateM	xhigh	+	Our developed profile	89 tasks, 445 trials	92.1% ; 88/89 coverage
GPT-5.6 Codex	Sol xhigh	+	Published reference	89 tasks, 445 trials	84.9%; coverage not reported
GPT-5.6 StateM	Sol xhigh	+	Frozen GPT profile; public submission	89 tasks, 445 trials	95.28% raw ; 89/89 coverage
GPT-5.6 Codex	Luna	+	Published reference	89 tasks, 445 trials	76.7%; coverage not reported
GPT-5.6 StateM	Luna	+	Frozen GPT profile	89 tasks, 445 trials	85.4% ; coverage not reported
DeepSeek-V4-Flash			Our baseline	89 tasks, standard time-out	82.7%
DeepSeek-V4-Flash StateM		+	Frozen GPT profile	89 tasks, standard time-out	82.0%
DeepSeek-V4-Flash StateM		+	Adapted profile	89 tasks, standard time-out	88.09% (392/445)
DeepSeek-V4-Flash StateM		+	Adapted profile	88-task common core	89.09% (392/440)
DeepSeek-V4-Flash StateM		+	Adapted profile; descriptive aggregate	89 tasks, one extended-timeout task	88.76% (395/445)

Table 4 Terminal-Bench 2.1 system-level results and transfer regimes Reference rows reproduce the stated comparator result; StateM rows report our runs. The descriptive DeepSeek aggregate replaces only the five `gpt2-codegolf` trials with the disclosed extended-timeout evaluation

Abstract failure mode	Generalized StateM control	Enforcement and evidence	Main applications
Phase and side-effect leakage	Separate read-only discovery and planning from execution. File edits and external mutations are permitted only after the task-derived plan is ready.	Workspace snapshots and read-only locks enforce the phase boundary; subsequent scope checks compare against the active task workspace.	RefactorBench, WebTest, side-effect workflows
Arithmetic and dependency drift	Route consequential arithmetic through exact-decimal tools, separate numeric inputs from narrative metadata, and reconcile derived values before acting.	Compact calculation templates record task-visible operands, operations, units, rounding, and independently checked outputs.	Budget approval, payroll, machine operations
Missing, duplicated, or partial effects	Use a task-derived effect manifest followed by execute and fresh-read reconciliation. Every mandatory effect must succeed exactly once before handoff.	Structured receipts compare declared, attempted, succeeded, failed, and freshly observed effects; optional effects remain explicitly distinguishable.	Budget approval, WooCommerce, WebArena
Cross-destination entity mismatch	Freeze one global task-derived entity set, then derive each destination’s write delta from its observed state and selected write mode.	Batch closure checks submitted, succeeded, failed, and observed entity IDs, cardinality, deduplication, and destination-required fields.	WooCommerce and other multi-sink workflows
Bulk-data distortion and context bloat	Keep tables and repeated records at the tool or file boundary instead of repeatedly transporting them through the dialogue context.	Batch tools consume structured inputs and return compact counts, aggregates, errors, and receipts rather than full payloads.	Machine operations, payroll, commerce workflows
Transient versus deterministic failure	Retry only failures classified as transient, with a bounded retry budget; do not retry contract, validation, or deterministic task failures unchanged.	Each retry uses fresh state where necessary and is followed by a public fresh-read check. Terminal errors remain explicit incomplete outcomes.	All tool-using families and benchmark runner
Service and session persistence	Ensure required services survive the agent command session and verify them through a fresh client or authenticated round trip.	Readiness depends on observable behavior, stable process ownership, and post-action state rather than fixed sleeps or shell-local process existence.	WebArena, WebTest, machine operations
Visible-contract and scope drift	Prioritize visible-contract fidelity and required reference closure; compatibility is preserved only with visible evidence, and minimality is a tie-breaker rather than the primary objective.	Read-only contract extraction, task-derived edit scope, focused public tests, and a final repository-wide stale-reference scan.	RefactorBench and repository maintenance
Ambiguous API migration	Prefer named keyword arguments when the visible contract names a parameter, and prefer direct symbol imports when consistent with the requested migration.	These are soft defaults. Positional-only APIs, cycles, lazy or optional imports, monkeypatching, module identity, and repository conventions override them when supported by public evidence.	RefactorBench and general code changes
Incorrect test or tool boundary	Keep inspection physically read-only and require authored tests to use the configured evaluator-free public runner rather than an incompatible host tool.	The workspace unlocks only on execute; host test entry points are shadowed, while public-runner results provide the verification receipt.	WebTest
Verification overhead and repeated gating	Generate templates and receipts at tool boundaries, advance state from validated tool events, and compress multiple checks into one evidence-bearing boundary.	Normal paths avoid repeated StateM round trips. A read-only reviewer runs only for elevated visible-contract risk and sufficient remaining time; unknown findings never block.	All families; reviewer used for RefactorBench

Table 5 Generalized Businessbench workflow controls distilled into StateM. Each control targets an abstract and reusable failure mode rather than a benchmark-specific answer. Configuration is derived only from the visible task, the active workspace, public documentation, and observable tool results. Hard gates are reserved for evidence-bearing invariants such as mandatory-effect completion, edit boundaries, and batch closure; preferences that may conflict with legitimate repository designs remain soft and overridable.